WOMEN E-GUARD

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This Report Presented in Partial Fulfillment of the Requirements for the Degree of Bachelor of Science in Computer Science and Engineering

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APPROVAL

This Project titled "**WOMEN E-GUARD"**, submitted by Monaf Hossain Fuad, Abu Bakar Siddique and Md.Moin Hossain, ID No: 151-15-4671, 151-15-5494 and 152-15-5761 to the Department of Computer Science and Engineering, Daffodil International University has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on 14th September 2019.

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DECLARATION

We hereby declare that, this project has been done by us under the supervision of **Zerin Nasrin Tumpa**, **Lecturer, Department of CSE** Daffodil International University. We also declare that neither this project nor any part of this project has been submitted elsewhere for award of any degree or diploma.

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ABSTRACT

Women harassment and violence has now become a big concern for us. The rate of women harassment and violence increasing day by day. Women do not feel free to move everywhere. We survey the women and found that most of them are being harassed and violated. In this digital world, we should take some digital step to make a secure environment for women. This document describes women safety and security by using electronic devices. By the help of GPS, women can get help from the people and make herself secure. By pressing her mobile phone button 5 times in a row, she can able to share her physical location and necessary information with the pre-selected contacts and police. Women can also be knowing about the nearby hospital and police stations. Most of the people now using the mobile phone that is android operated. So, by using the android platform and with the help of the Internet and GPS, now we can secure our women. We focused on the proposed model that can be used to deals with the problem of security of women using the Internet and Global Positioning System (GPS).

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CHAPTER 1

INTRODUCTION

1.1 Objective

The main objective of this project is to develop an android application to ensure our women safety. This is the solution of the security of common women harassment. The user will be able to share her location with her family members, friend and security forces. She can be able to getting help from nearest volunteer (pre-registered). Victim's family members, friends and security forces will be able to check the position and also can be able to listen what is happening there and can try to contact her through this application. Volunteer can be able to see the location and can go for rescue. Another objective of this project is to explore the field of android application in security system and relevant things.

1.2 Motivation

Android is one of the most popular and common operating system for mobile phone. In Bangladesh most of the mobile phone user are using android operated handset. There are millions of android application for different platforms. On the other hand women harassment is risen day by day in Bangladesh. So, in the field of stopping women harassment and for making a women secure society we can use our smart phone. User will be able to get help from the good people and make herself secure from the bad people. User will be able to share with location with her family, friends and law force department. Global positioning system is used here to navigate victim.

1.3 Problem Statement

Most of the woman of our country everyday facing sexual harassment in verbal, physical and other forms. About 94 percentage of the women of Bangladesh facing these types of harassment those who are commuting in public transport [1]. Around 84% of Bangladeshi women are constantly being subjected to harassment in public, streets, vehicles, markets or at home [2].

41 percentage of women are facing sexual or non-sexual harassment in public transport [2]. From our online survey we know that these incidents are hindrance to women's development.



Figure 1.1: Harassment of women [3]

1.4 Expected Outcome

Our main concern is to make a women-friendly environment, where women can secure from the violence. Our women will feel free to move ahead. Women can feel safe in our public places. Percentage of women harassment will decrease and participation in development work will be rise. Finally, we can make a society where women can feel secure and feel free to fly.

1.5 Report Layout

The entire project is composed five chapter. In the report, the layout is summarized that six chapter. Discuss the summarized below:

- In chapter one gives an introduction to Women E-guard about android application and its motivation, objective and overview of whole system.
- In chapter two covers an extensive literature review of previous works on Women Security application, its system design and similar others work and show technical Challenges.
- In chapter three highlights the project use case design with the description, block diagram, details about our proposal and our implementation requirements.
- In chapter four of the project design and implementation with practical details of our project, research, and implementation for testing the project.
- In chapter five total testing and deployment process has mentioned.
- In chapter six is on the conclusion and recommendations based on the challenges counted and future possible development of the project work are counted.

CHAPTER 2

LITERATURE REVIEW

2.1 Android an Introduction

Android is one of the most famous mobile operating gadgets of the cutting-edge technology. Android is a very easy and smart operating system. And it is up to date day by way of day. Android utility makes our lifestyles faster and easier. It is a Linux based running gadget it is designed in particular for contact screen cellular devices such as clever cellphone and pill computers. The android is a powerful operating gadget and it helps large quantity of applications in Smartphones. The android has received tens of millions of apps handy that can assist you managing your life one or other way and it is accessible low fee in market at that motive's android is very popular.



Figure 2.1: Logo of Android [4]

2.2 Android Application in Women Security

Android application collect the necessary information like location, position, time and store it on the server. The server sends the data to user's selected persons and security forces. The full procedures happen through the internet. Here android application works through the internet and makes the connection with their trusted people and security forces.

2.3 Literature Review

Harassment covers a wide vary of offensive or unwanted conduct in order to make other person uncomfortable. Most regularly female are the victims of harassment and the offenders are male. Harassment is a structure of illegal and discrimination and all the human rights legal guidelines restrict this act but ratio of the harassment is always growing although legal guidelines are current in our society (Harassment of Women, 2018). In the Bangladesh female harassment is growing very rapidly. Every day we located the information about girls' harassment.

We survey the women from the various sector and found that the maximum has a bad experience of women harassment. In our survey, we found 91.2% of women is travel along and 81.4% of them use public transport. 64.5% of women are faced with harassment. 96.5% of women ensuring us that women harassment is increasing nowadays. From our survey, we also know that 99.1% of women feel that women harassment is a hindrance to women's development. And 100% of women, who are involved in our survey believe that women will be developed if they get a secure environment. [5]

2.4 Existing System

In Bangladesh:

JOY (App to combat violence against women and children):

The mobile based JOY application software can help women and children in emergency situation whenever they face violence. Victim can get the help by pressing emergency button on the app, which will send text messages to the police super of the area, metropolitan deputy

police commissioner, three pre-selected ‘friends and family’ numbers, and to a national help center (109). [6]

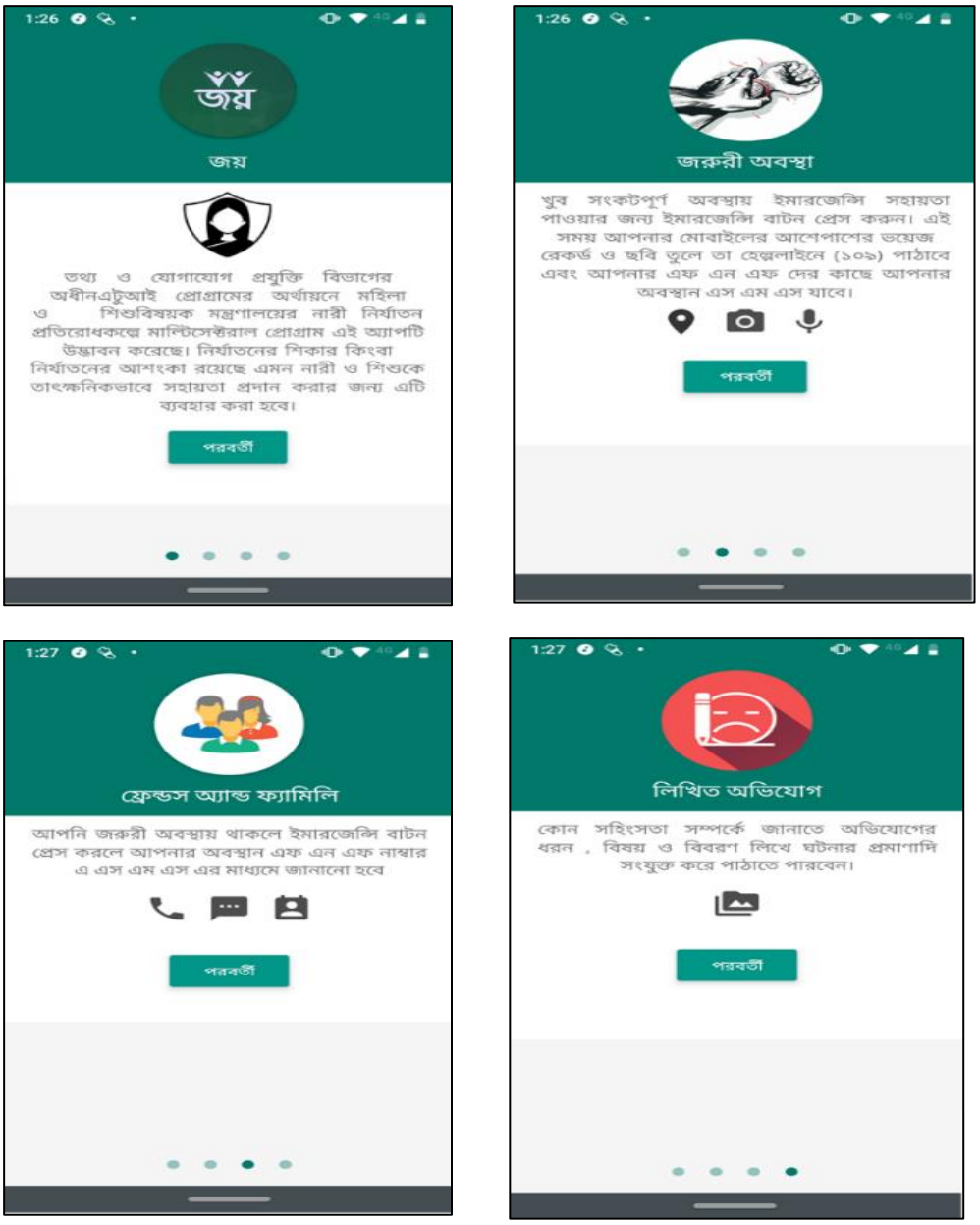


Figure 2.2: Joy app preview [7]

But these app is not working popularly because of its limitation. On the other hand this application needs National Identity number for using it. So, teenagers are not able to use this app. showing in fig. 2.4.2.

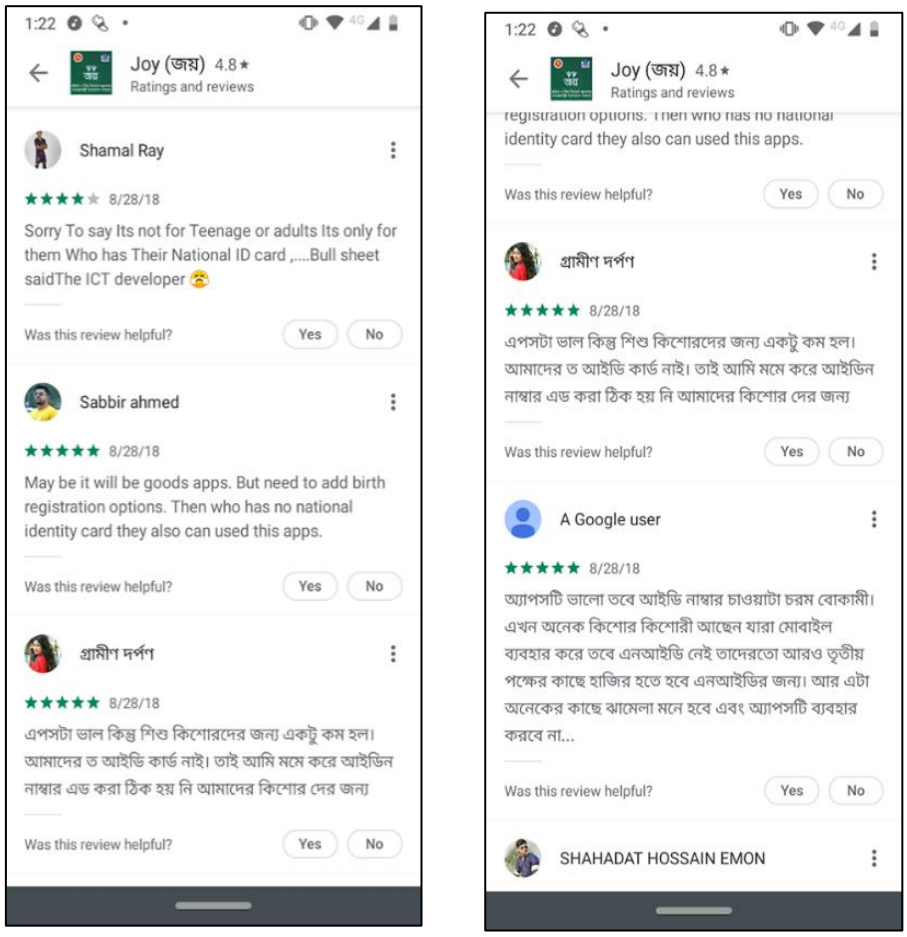


Figure 2.3: Joy app review [8]

In Global:

1. VituU App

This app start working by pressing the power button twice. Whenever, the victim **press the power button twice**, the system send and SOS alert to contacts instantly. Alert messages are sent to the listed contact in every two minutes with the victims' physical location.

Now, this application is removed from the google play store, because of its system fault. The twice press of power button makes this system so annoying. People are pressing power button unintentionally and this app started working. [9]

2. Circle of 6 App

This app makes communication between a circle of 6 people (user pre-selected) and user. By the help of this app, the user can send a message, make a call and request her circle for taking her home or come to the place where she is.

Now this app is no longer on play store. [9]

3. Life360 Family Locator App

By sharing the location of user among the family members and friends and keep a handle on her (user) location, the Life360 app is working. The circle members can chat with the user and can get updated with the user physical location every minute. [9]

4. I'm Shakti App

Pressing the **power button 5 times in 2 seconds** then the app is triggered. This app shares the user location to her emergency contact people by SMS.

This app is only for the Indian users, Bangladeshi people are not able to use this app. [9]

5. Famy Family Chat & Locator App

This app is helpful to the parents who have a feisty kid, or one who has to watch over the teenage siblings who have an issue with "personal space", this may be the locator app the people need. This app picks up the exact location and shares the location to the group. This app keeps track of all the previous locations and will let users send an SOS alert when necessary.

Now, this app is unavailable from the google play store. [9]

6. Nirbhaya: Be Fearless App

User can able to send an SMS alert or call with a single touch in the event of an emergency by using the Nirbhaya app. When the app is activated, it will send the user's GPS location to her contacts (pre-selected). This app has "**Shake to alert**" function and user can also use the power button to send an SOS alert to the pre-selected contacts.

This app is only for the Indian users, other country people are unable to use this app. [9]

7. Watch Over Me App

The watch over me app helps the user to share her journey location with her chosen contacts. This app sends an emergency alert to the chosen contacts when the user fails to check-in from the intended location. Alternatively, in case of a bad situation, the user can **shake her phone to trigger an emergency alert** and the user's **device camera will turn on automatically** and **start recording**. This app also notified the user about high crime rate area.

This app is now available in 7 cities of India only.

8. Sentinel Personal Security SOS App

This app claims to be the fastest SOS app available in the current app market. This app sends SMS and email alerts, as well as call users emergency contacts when the need arises. The SMS alert contains GPS location, time and direction of travel. **Sentinel Personal Security SOS App** works even without a working Internet connection or network. [10]
This is a paid app and this is not working in the all android updates.

9. Secure Her App

Secure Her App works by sending out a distress signal as soon as the phone user **double taps on the app's icon**. Then a **Call Centre will respond** to the distress signal to make sure that help reaches the user.

Currently, this app is only works in India. [11]

10. Women Safety Shield App

Women Safety Shield app allows to **take pictures (as evidence)**, and quickly send the current location to the users pre-determined emergency contact. There's also a function called "**Walk With Me"** that lets user-selected contacts follow her travels to keep tracking. This app also provides a list of nearest **hospitals and police stations**.
This app is now only available for the users in India.

11. iGOSafely

Whenever the user needs to get help, this app is activated by the shaking of the user's phone. Then the app sends alert messages, street address, emails, and GPS position to her emergency contacts to make possible actions. It also sends a 30 seconds audio records in every 60 seconds which is activated by the shaking the phone.

This app starts working when the headphone cord is removed from the phone and it makes people so confused. The beta version of this application is running on the market, but it is unavailable for Bangladesh.

12. Shake2Safety

The android app Shake2Safety sincerely achieves the core purpose: Women's complete safety. Upon encountering a precarious situation, the app sends an SOS message to preconfigured contacts once you shaking the phone or press the power button four times.

Phone shaking option is subject to user preference and comes handy when you need emergency help but are unable to operate the device.

This app is currently unavailable from the google play store. [11]

13. Women Safety

Along with sending an instant message to the emergency numbers, it offers an additional service of sending an email to pre-configured email IDs of user trusted contacts with location details and Google Map link.

Apart from the usual, the app adds more value and takes two pictures from the phone's front and back camera as well as a supporting video and audio clip and sends them all to user contacts via emails.

This app has some technical issues like this app couldn't work when the phone is locked. There is no edit option to chance any email or number. [12]

14. CitizenCop

CitizenCop is quite an ideal app for ensuring women's safety by reporting criminal incidents at the earliest. If a woman is caught in a dangerous emergency or sees anyone facing a situation of peril, she can use CitizenCop to report it immediately. The app is thoughtful about women's safety concerns and makes the job of communicating crime to the law enforcement department easy and rapid.

This app has some problem with the phone tracking and sometimes it doesn't give any update.

15. Family Locator – GPS Tracker

The app uses a device's in-built GPS sensors to inform family members whenever any of them checks in anywhere. The notification is sent to everybody once the member safely reaches the pinned location on the map.

This app could not work in the background. Sometimes this app couldn't track the exact location.

16. Eyewatch SOS for Women

Eyewatch SOS for Women is an excellent app focused around the safety of today's women. It sends the audio and video of the user's surroundings to the emergency contacts along with an alert message.

When user reach the location safely, user can press the apps **I am Safe** button to inform your folks of your safe arrival.
This app makes devices unusable sometimes and the phone would start dying quickly.
Battery huge useless. [12]

17. My SafetyPin

My SafetyPin acts as a smart companionable guide especially when the user is stuck at a crossroad and needs to choose the best optimal and safe route to your destination. If the user is caught in an unfamiliar or unsafe location, the app sends alerts to family friends who would track the user.

Whether or not the given area is safe is determined based on real-world factors including public transport, traffic conditions, general security, visibility, and other crucial elements. [11]

18. Raksha

Raksha is working by providing the user's current location with the selected contacts. Family members and friends can able to trace the user in trouble in the shortest time, whenever the user selects the "**Are you in trouble?**" option.

19. Watch.Me

This app will keep the users family and friends informed of the users' current location and can also share the live videos with selected emergency contacts.

User can pick any video to share including daily routine, personal trips and snap stories, etc.

20. bSafe

bSafe is one of the best women's safety apps that provides SOS service in a situation of distress to send the message along with the GPS location to the emergency network of guardians which user can create.

User can also record and send the video and audio to the group of emergency contacts. [12]

21. Himmat Plus

Himmat Plus was originally launched by the Delhi Police of India and is highly recommended for women facing unpredictable issues and emergencies especially when they move around alone. Marked with a user-friendly interface, the app is easy to use, as easy as simply sending the OTP on the smartphone for user verification after it is installed. [12]

The moment user feels their safety is compromised, they select the panic button so that their precise location is immediately shared with Delhi Police unit along with audio or video content. [13]

2.5 Problems in Existing System

There is a lot of research and development going on android applications. But in the case of women security don’t full fill the requirement of women security. The features of the existing system are not being high. Some of them have a lack of system. Most of the existing application is not working properly.

Chapter 3

REQUIREMENT & SPECIFICATION

In this segment we made those documents that will capture the complete information about how our app will work, what it will do. REQUIREMENT SPECIFICATIONS is one of the most crucial part in our project.

3.1 Business Process Model

Commerce handle modeling program construct approach to speak to your handle in a advanced energetic way that can be moved to a live machine-driven strategy. The term "business model" is utilized in exceptionally recognizable way and for a expansive extend of casual depictions to speak to center perspectives of counting reason, structures, operational handle and approaches. The most characteristics of the technique depends on graph as "Flow diagram". Here I'm endeavoring to clarify my ventures commerce demonstrate utilizing Information Stream Graph. Regularly, Information Stream Graph depict how information stream is dealt with through a framework or venture. A Information Stream Chart appears what sensibly information are input to and yield from the framework. How the information will prepare through the framework and where the information will be spared. It does not appear data approximately handle timing or whether forms will work in arrangement or in parallel, not at all like a conventional structure flowchart which centered on control stream, it's seems like as diagram.

By plan of action client will see how it will function and that is the central matter. Any place it utilizes and any place its utilization case. A designer consistently attempts to make his/her plan of action by his/her own specific manner. Because of every application is select that is recommends that unique with each other. The accompanying figure 3.1 demonstrates the business procedure model.

User can add emergency contact numbers. Those numbers can be edited or deleted if the user wants to.

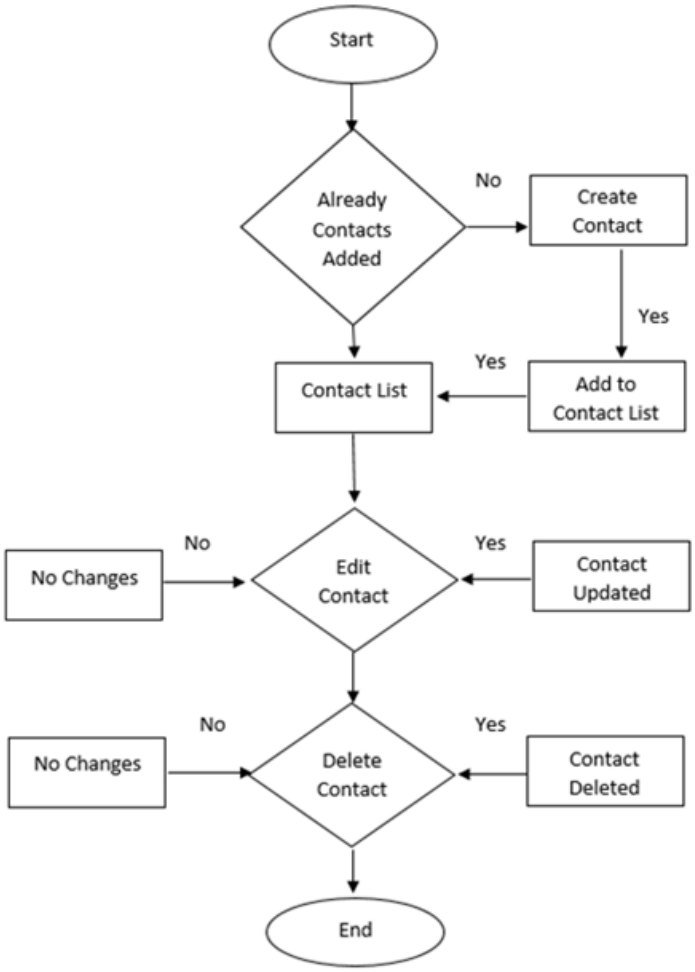


Figure 3.1: Shows the Data Flow Diagram of Women E-guard

3.2 Requirement Collection and Analysis

This undertaking has several varieties of necessities. It's fundamental to spot the easiest necessities required for this task and style the modules in such way that all the demands can be satisfied. We work step with the aid of step. This step needs a radical investigation of the necessary needs for the satisfaction and conjointly the problems encountered. We acquire various data from one of a kind type of social internet site. We start to work on our venture

based totally on this data. Only the introduction would be adequate to furnish a extra strong performance.

3.3 Functional Requirement

Functional requirement is interconnected to the specialized utilitarian of the framework as application programming can perform. From the perspective of our framework, it ought to have a few practical needs like looking after database, get area facilitate, send SMS to crisis contacts, and get closest police headquarters area.

3.4 Non-functional Requirement

Non-functional request is a necessity that indicates criteria that may likewise use to decide the activity of a machine particularly conditions, as a substitute of explicit practices as simply like the utility is the thing that amount proficient, individual agreeable, in general execution issue of the application and so forth. From the perspective of our framework, the apparatus has incalculable reasonable necessities as a ton of productive, important, advance execution, memory purchaser and smoother activity. The User interface of utility is buyer wonderful and it's amazing for client experience. Non-practical various join work moreover uses in here. This product can be utilized exclusively on android OS. No one but individual can modify the data of the application. An android OS based machine with SIM appended and GPS support.

3.5 Use Case Diagram

Use case is a rationalization of how a person can use the gadget to gain his enterprise goals. We understand that use case is that case that the usage of all running method in step by step. In software and systems engineering, a use case ought to be an all listing of movements or tournament steps, usually make clear the interactions between a system, to reap a goal. Use-case constantly use for user perception and that I tried my most for efficiency.

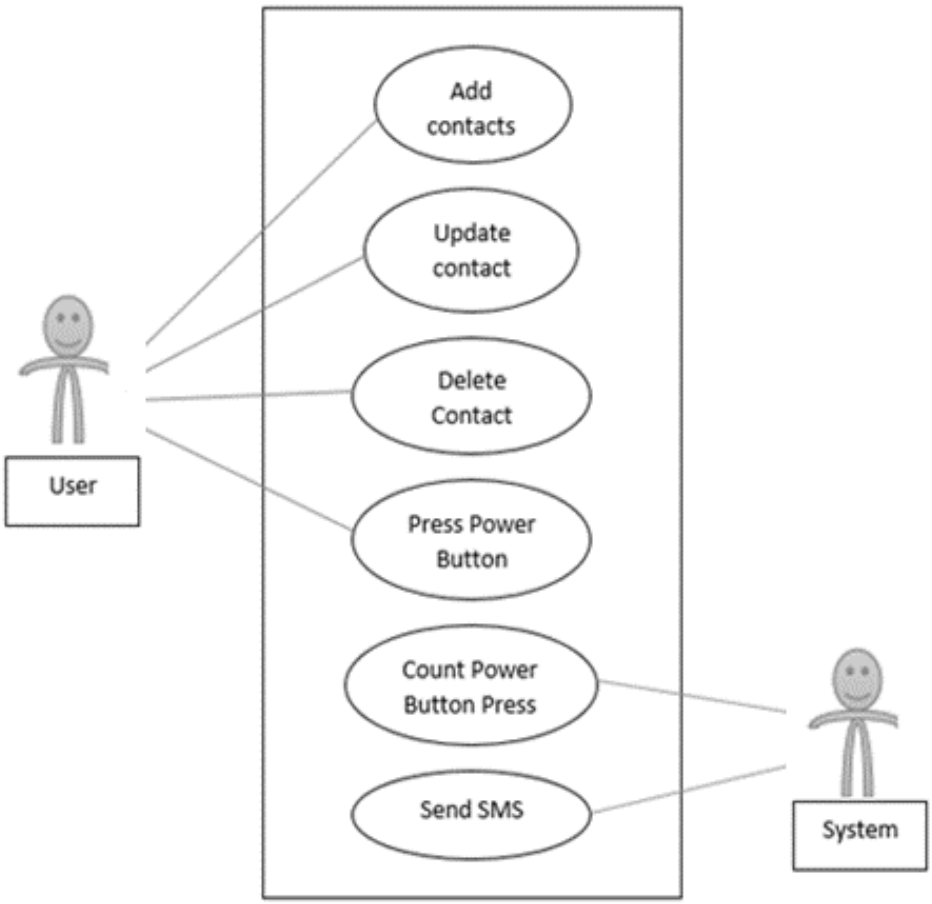


Figure 3.2: Use-case modeling of Women E-guard

3.6 Logical Data Model

A logical data model is a information model of a chose issue space conveyed unreservedly of a specific database thing or limit development in any case as to such an extent as data structures, for instance, contact list. As a Back-end I use core object oriented and Front-end XML. In our application there are database section and SQLite as database server. Our application needs to store contacts number so, that's why there is database part in the application.

3.7 Design Requirements

The try to build our app as simple as possible. We always think about user when we are making this app. We attempted our best to build our application easy to use.

Here some requirement:

1. In our app when user launch the app the first activity will ask user to give permission for to access GPS service and send SMS.
2. After that there is an activity where user can add contacts number.
3. In our app there is no login system because in emergency if user wants to add some new contacts, there no need to time waste by typing user login information.
4. Only user can add, modify, and delete app data.
5. Simple user interface, no complex user interface in the app.
6. Need an android phone which have a sim attached and have GPS service.

Chapter 4

DESIGN SPECIFICATION

In this Design Specification chapter will going to discuss how we design all front-end, back-end design and collaborate design. We will show the design of our app. We use java and XML to build our app.

4.1 Front-end Design

A front-end is graphical user interface by which user can communicate with back-end without any knowledge about it. It's help user to use the app easily. If front-end of an application is simple and user-friendly than user like to use app, but if front-end isn't simple or complex user don't want to use that app again. So, we tried to build our application front-end user interface as easy as possible.

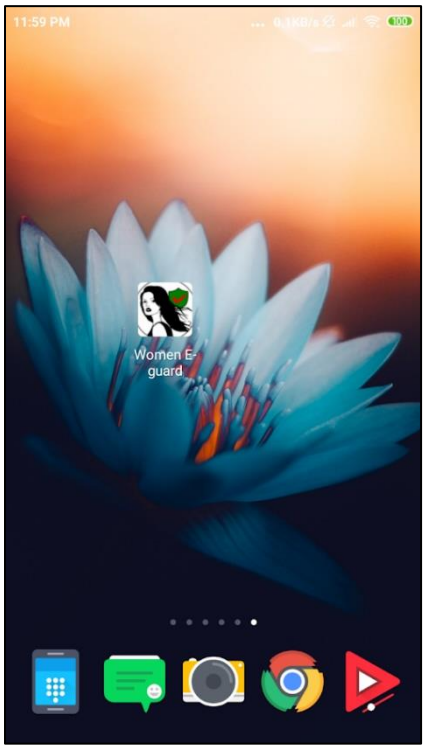


Figure 4.1: Screenshot of Women E-guard Icon

We try make an icon by which user can easily understand what kind of application it is, how it will work. By watching this icon symbol user can easily identify.

Grant permission:

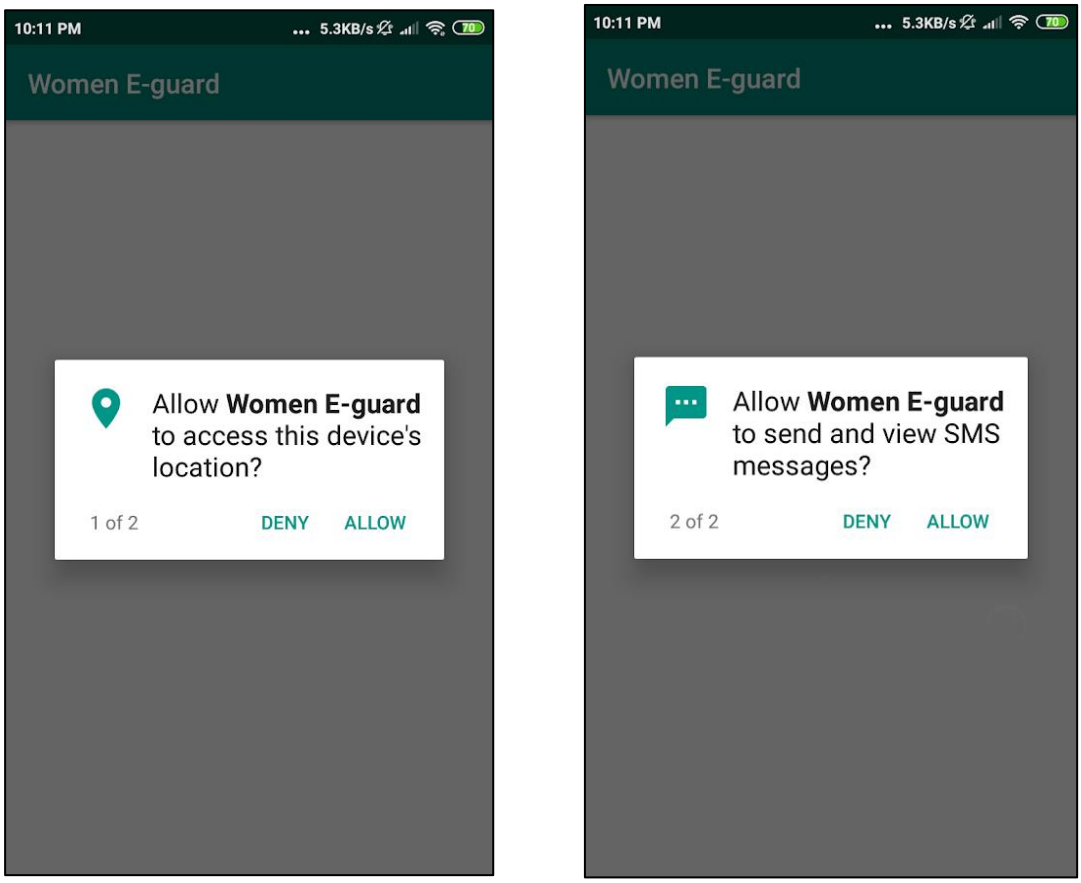


Figure 4.2: Screenshot of asking permission.

User have to allow above in order to use our application perfectly. There will be two kinds of permission our application will ask. First one is for device's location. After user allow access device location permission our application can use GPS system of the device. By using GPS application can get the coordinate. Now, second one is SMS messages permission our app will need. By granting this our app can send text messages to emergency contacts number.

Add first Number:

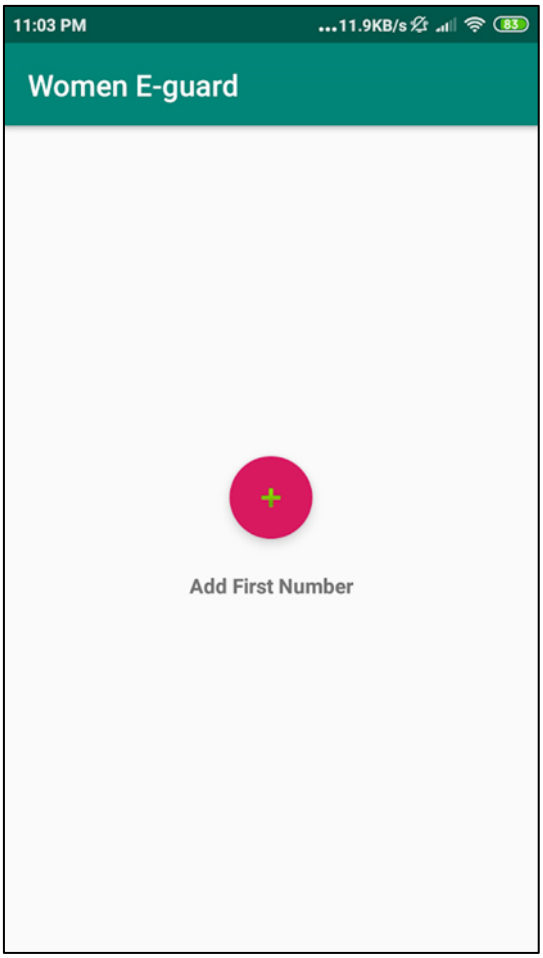


Figure 4.3: Screenshot of Add first number activity.

After granting device location and SMS permission, app will open this activity. So that user can add contact name and number in our app. By pressing circle pink button in fig. 4.3 user can add contact info.

Save Contact:

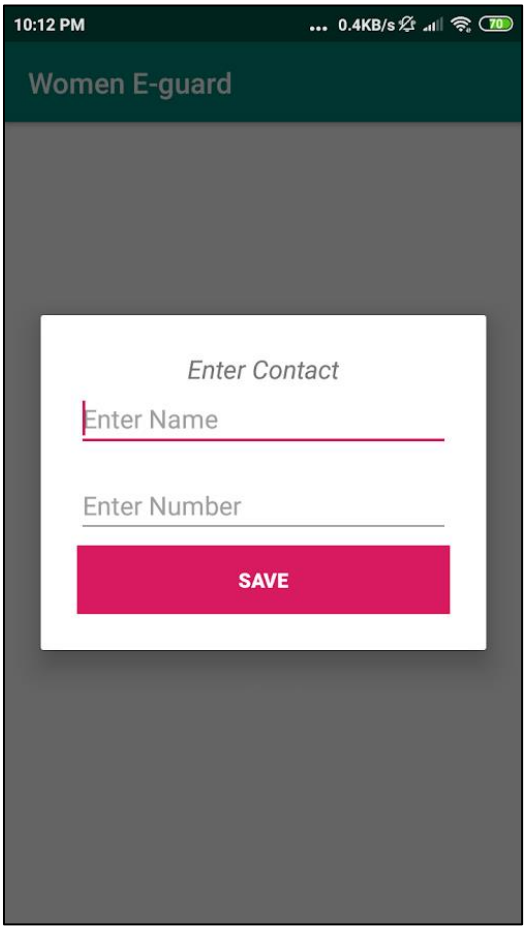


Figure 4.4: Screenshot of save name and number.

After pressing circle pink button from fig. 4.4 above window will be open. In this window user will add emergency contact name and number to save it into application (application will store contact info in SQLite database).

Contact List page:

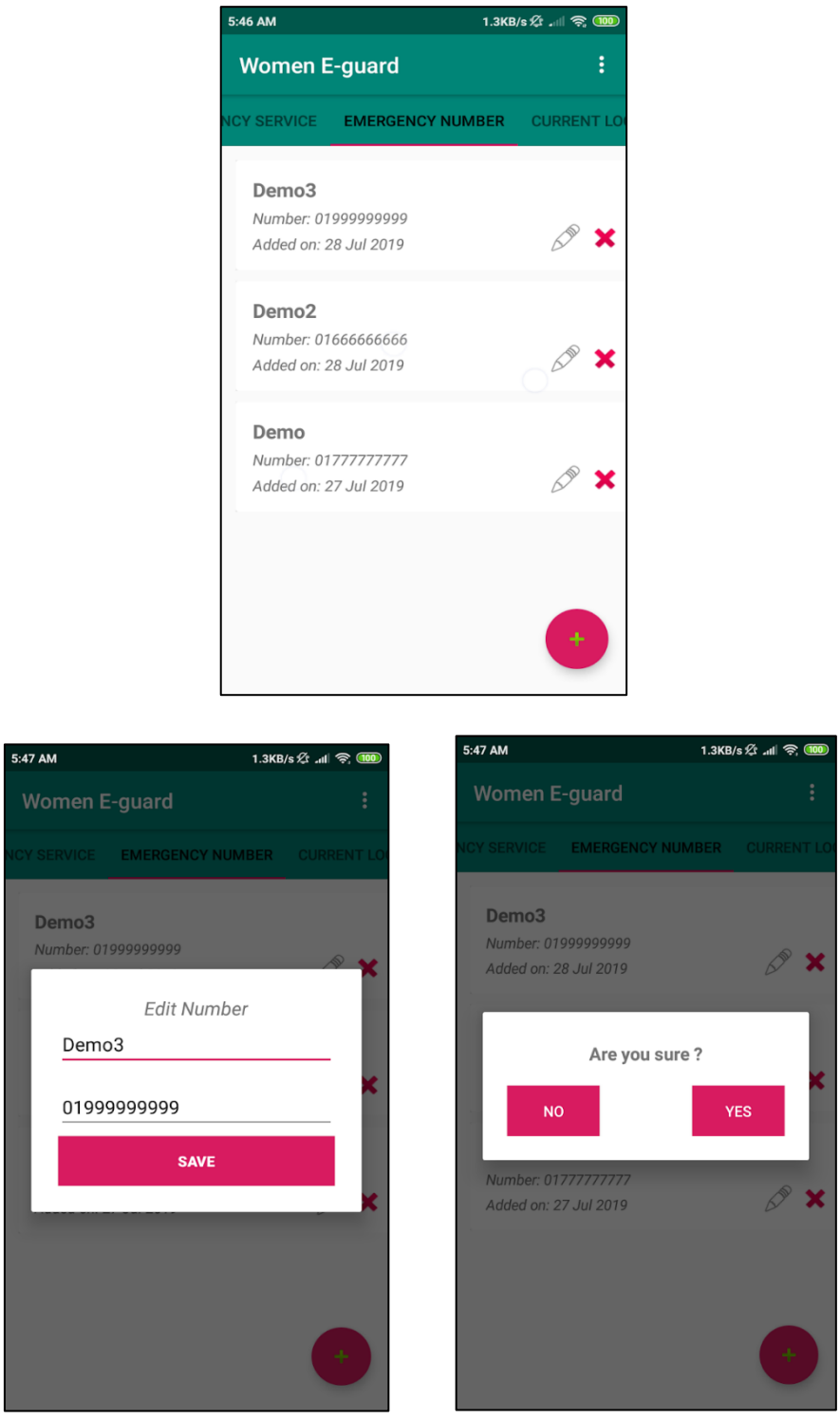


Figure 4.5: Screenshot of number list (left), Edit contact window (middle), delete contact window (right).

In above figure user can show contact number list in new to old order. If they want to edit any contact information, they have to press pencil button and it will open a new window where user can edit contact information. There is cross like another button in the number list. By tapping that button user can delete contact. If user press that cross button a new confirmation window will open. So, user can confirm that the selected contact will be deleted or not.

Menu Bar:

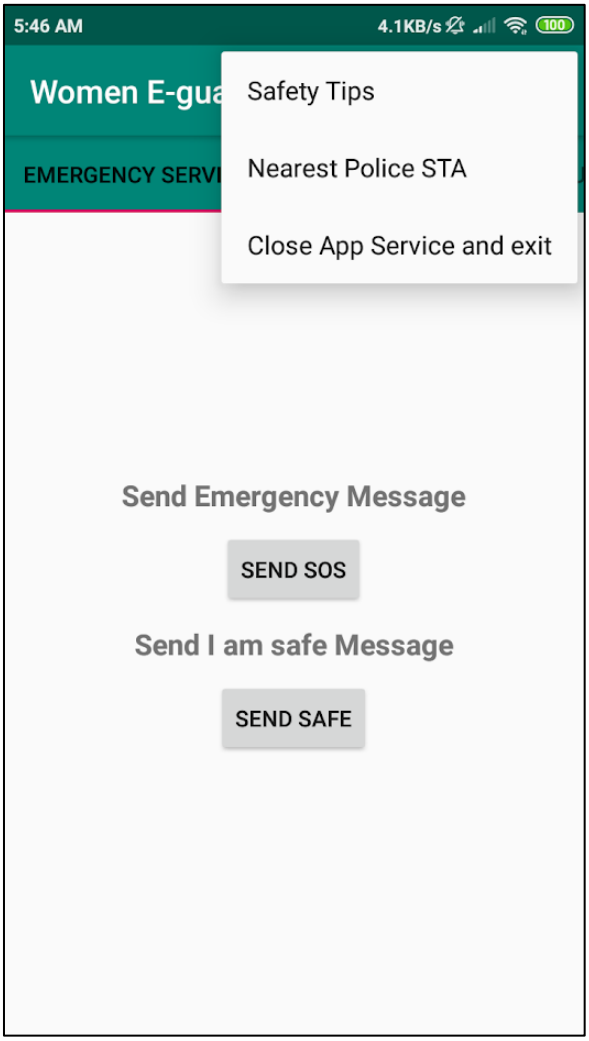


Figure 4.6: Screenshot of menu bar.

There are two options in the menu bar. First one is safety tips and second one Nearest Police Station.

If user tap safety tips option from menu bar then it will open women safety tips page. And again, if user tap Nearest Police Station option from menu bar then will open google maps application to user device and it will show user the nearest police station from user current location.

Safety tips page:

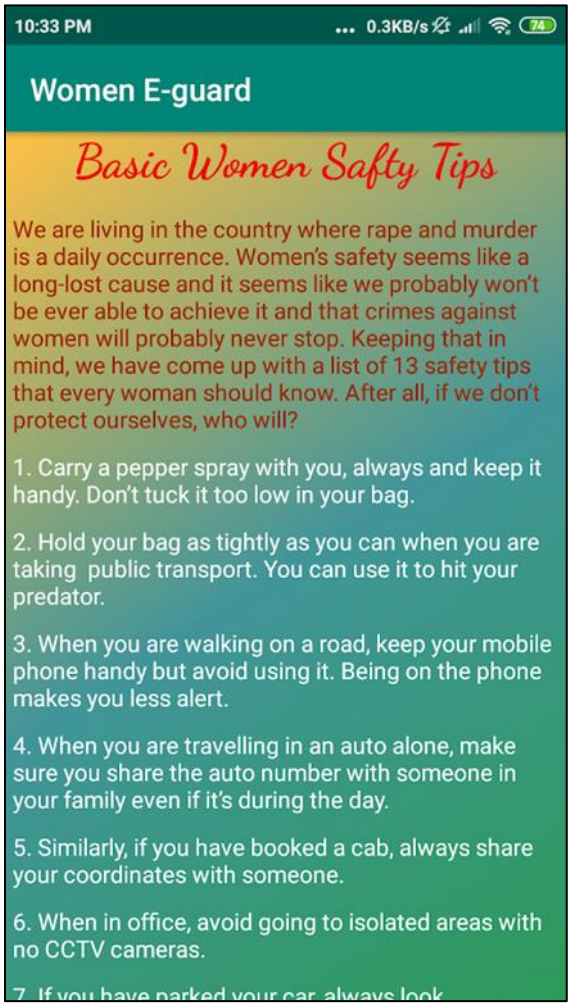


Figure 4.7: Screenshot of Women safety tips.

By open this activity user can read and know about basic safety tips how user can keep herself safe.

Nearest Police Location:

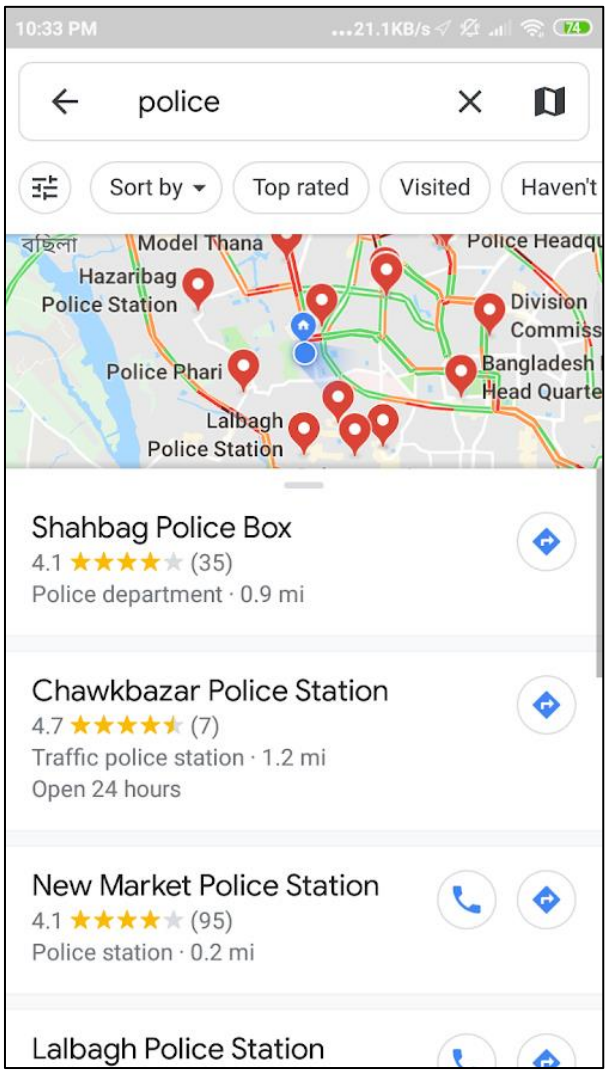


Figure 4.8: Screenshot of google maps app showing nearest police station.

From fig. 4.6 there is an option in menu bar called Nearest Police Station. When user tap that option, it will open google maps application where user can get information about which police station is nearest from user current location.

4.2 Implementation Requirements

Every application need implementation tools to build. We used programming language such as java, XLM, Android studio, valid phone number to send message, an android phone to use our application, google account (google maps app need google account to work).

Chapter 5

IMPLEMENTATION & TESTING

5.1 Implementation of Database

Database use to hold user information into the system. Implementation of database is the method of installation database system. We use SQLite database system in our app to save user data. SQLite is best for offline based approach. That's why we used it on our app.

Table 5.1: Database Table for Women E-guard

Id	Name	Number	Date
1	Demo1	017xxx	hh:mm:ss
2	Demo2	015xxx	hh:mm:ss

```
//CRUD operation: Create, Read, Update, Delete

@Override
public void onCreate(SQLiteDatabase sqLiteDatabase) {
    String CREATE_TABLE = "CREATE TABLE " + Constants.TABLE_NAME + "("
            + Constants.KEY_ID + " INTEGER PRIMARY KEY," + Constants.KEY_NAME_ITEM + " TEXT,"
            + Constants.KEY_NUMBER + " TEXT, "
            + Constants.KEY_DATE_NAME + " LONG);";

    sqLiteDatabase.execSQL(CREATE_TABLE);

}
```

Fig 5.1: Database design of Women E-guard

5.2 Implementation of Front-end Design

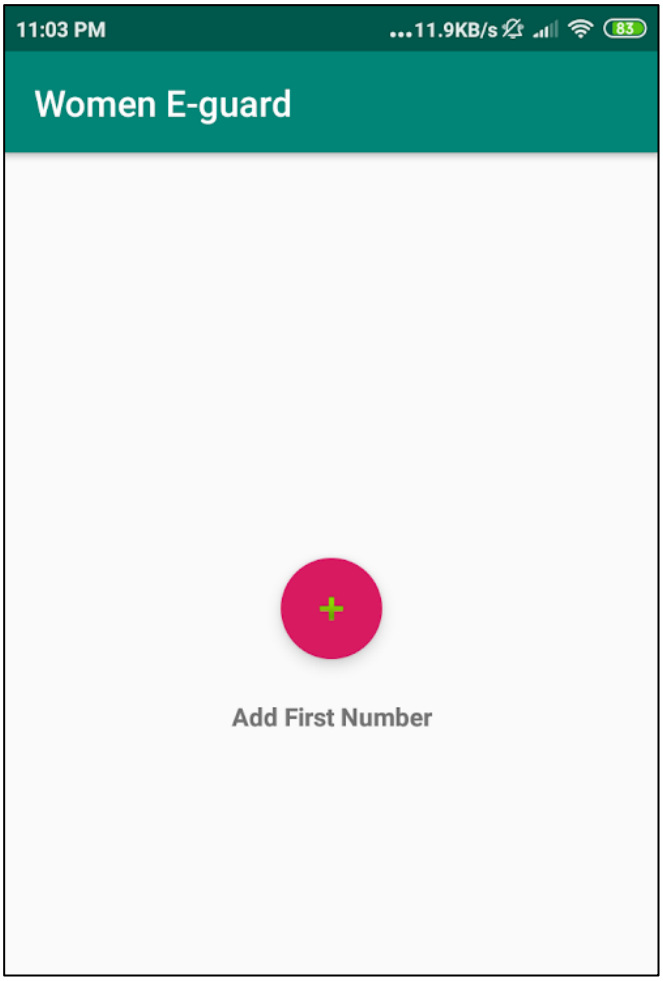


Figure 5.2: Main Activity page

Women E-guard system front design build up by using below items-

Language: Java, XML

For Design: Adobe Illustrator

Platform: Android Studio

5.3 Implementation of Interaction

Our app is fully implemented for user. We build our app as easy and simple as possible. So, that user who will use this app can easily use it without facing any problem. We implement our app whit responsive user interface for grate user experience.

5.4 Test Results and Reports

Test Case	Test Input	Expected Output	Actual Output	Result
1. Open the application	Uses some android OS phones	Run the application successfully	Run the application successfully	Passed
2. Add number without name	Blank or incorrect input	Contact details added	Please input name and number	Passed
3. Add name without number	Blank or incorrect input	Contact details added	Please input name and number	Passed
4. Add name and number	Input name and number	Contact details added	Contact details added	Passed

Table 5.2: Test Case Table for Women E-guard.

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 Discussion and Conclusion

The objective of this project is to make an android app for women security. There was also another objective of the project is to experiment with the android app in real life. As this technology was new us it was a very big challenge to cope with this field. We hope we will develop this project adding more feature and make it more interactive and smarter. Our only aim is to make secure Bangladesh for our women. We strongly believe that, if we make a safe environment for our sisters, their participation in the development work will rise.

6.2 Limitations

Our app can't work without the internet, sometimes user don't have data to using this application.

6.3 Future Scope

- Worldwide use
- Smart Button (wrist watch and locket)
- Face detection using Open user's device camera
- Internet free android app
- Conversation between police and user remotely

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